

Jackfruit Phase-1

Software Requirements Specification

**For EV Charging Station Finder with Slot Dashboard and
Auto-Booking System**
Version 1.0 approved

Team details

Team Name	Project ID selected	Team Member 1 name and SRN per PES records	Team Member 2 name and SRN per PES records	Team Member 3 name and SRN per PES records	Team Member 4 name and SRN per PES records
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Revision History

Name	Date	Reason For Changes	Version
Vaishnavi Singh, Atharva Ravi Jamakhandi, Velkur Tanisha Reddy, Vaibhav Tulsian	5th September 2026	Initial draft	1.0

This SRS covers the complete system, including all four subsystems:

- User Accounts, Vehicle Profile and Access Control – Vaibhav Tulsian (PES2UG24AM176)
- Station Discovery, Navigation and Real-Time Slot Dashboard – Atharva Ravi Jamakhandi (PES2UG24AM196)
- Battery-Aware Booking and Auto-Booking – Velkur Tanisha Reddy (PES2UG24AM181)
- Payments, Billing and Usage Analytics – Vaishnavi Singh (PES2UG24AM177)

1. Introduction

1.1 Purpose

This document specifies the software requirements for ChargeSlot, version 1.0, a mobile application that helps electric vehicle (EV) drivers find nearby charging stations, see which charging slots are free in real time, and book a slot whose length is matched to the battery level of their vehicle. The product also provides navigation to the station, in-app payment, and usage analytics. It is the working name for the “EV Charging Station Finder with Slot Dashboard and Auto-Booking System”.

This SRS covers the complete system, including all four subsystems:

- User Accounts, Vehicle Profile and Access Control
- Station Discovery, Navigation and Real-Time Slot Dashboard
- Battery-Aware Booking and Auto-Booking
- Payments, Billing and Usage Analytics

It defines what the system must do. It does not define how the system will be built. Design decisions are covered in the separate Architecture and Design documents. The document follows the structure of IEEE Std 830-1998.

1.2 Intended Audience and Reading Suggestions

This document is written for the following readers.

Reader	What to read	Why
Development team	All sections	Full specification of what must be built
Faculty evaluator	Sections 1, 2, 5, 6	Scope, features, and non-functional requirements
Testers	Sections 5, 6, Appendix C	Functional requirements and traceability to test cases
Project manager	Sections 1, 2, 5	Scope and feature breakdown for planning
New reader	Sections 1, 2, then Appendix A	Overview first, then terminology

Readers unfamiliar with EV charging should read Appendix A before Section 5, because Section 5 uses terms such as slot, connector, battery level, usable range, and auto-booking with precise meanings.

1.3 Product Scope

What the product is. ChargeSlot is a mobile application, backed by a server, that tells an EV driver where to charge, when a charger will be free, and books it for them for exactly the time their battery needs.

The problem it solves. EV drivers do not know in advance whether a charger is free, working, and compatible with their car. They arrive to find the charger occupied, faulty, or the wrong connector type, and they lose time and may run low on charge. Each charging network has its own app and its own payment method. A driver also has no easy way to know how long their car needs to charge, so they either block a charger for longer than necessary or leave before the battery is at the level they need.

How it works. The driver saves a vehicle profile. The system receives live status from charging stations and keeps a dashboard of which connectors and time slots are free. The driver enters the current battery level and the level they want to reach. The system calculates how far the car can still

travel, which stations are reachable, how much energy is needed, and how long charging will take on each compatible connector. It then recommends slots, or books the best one automatically (auto-booking). The driver navigates to the station, starts the session from the app, pays in the app, and later sees usage analytics.

Objectives:

- Reduce the time a driver spends finding a working, compatible charger.
- Ensure a driver is never offered a connector that does not fit their vehicle, or a station they cannot reach with the battery they have.
- Size every booking to the battery need, so that chargers are not blocked for longer than necessary.
- Show real-time availability, so that the driver can see the current state of every connector.
- Give station operators visibility into how their chargers are used, so they can plan tariffs and maintenance.
- Show the driver the estimated cost before booking and make every recommendation explainable.

Out of scope for version 1.0:

- Manufacturing, installing, or controlling charging hardware. The system only sends the standard start and stop commands through the charge point gateway.
- Reading the battery level automatically from the vehicle. In version 1.0 the driver enters the battery level.
- Vehicle-to-grid, battery swapping, and home or residential charging.
- Payment of operator payouts. The system calculates earnings but does not transfer money to operators.
- In-app voice turn-by-turn guidance. Navigation is handed to the device's navigation application.
- Web application for drivers, and languages other than English.

1.4 References

Ref	Title	Version	Source
R1	Jackfruit Software Engineering Mini-Project Guidelines 2026, UE24CS341A	2026	Course handout
R2	IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications	1998	IEEE
R3	EV Charging Station Finder with Slot Dashboard and Auto-Booking System: Problem Statement (Project ID 37)	1.0	Course handout / Team submission
R4	ChargeSlot High Level Architecture Document	1.0	To be produced
R5	ChargeSlot Validation Specification	1.0	To be produced
R6	Open Charge Point Protocol (OCPP) 1.6 Specification	1.6	Open Charge Alliance
R7	Google Maps Platform Documentation (Maps SDK, Directions API)	Current	Google Developers
R8	Payment Gateway API Documentation (Razorpay, test mode)	Current	Razorpay
R9	Firebase Cloud Messaging Documentation	Current	Google Firebase
R10	Digital Personal Data Protection Act, 2023	2023	Government of India

2. Overall Description

2.1 Product Perspective

ChargeSlot is a new, self-contained product. It is not a replacement for an existing system and not a member of an existing product family.

It sits between three things that already exist but are not connected:

- The network of charging stations, whose chargers report their own status through a standard protocol (OCPP) but have no booking facility visible to the driver.
- The driver's vehicle, which has a battery size, a charging rate and a connector type that decide which chargers it can use and how long it needs.
- Map and payment services, which know where things are and how to move money, but know nothing about chargers or batteries.

ChargeSlot is the layer that connects them by turning the charger's status, the vehicle's battery and the driver's destination into a single booking.

System context diagram:

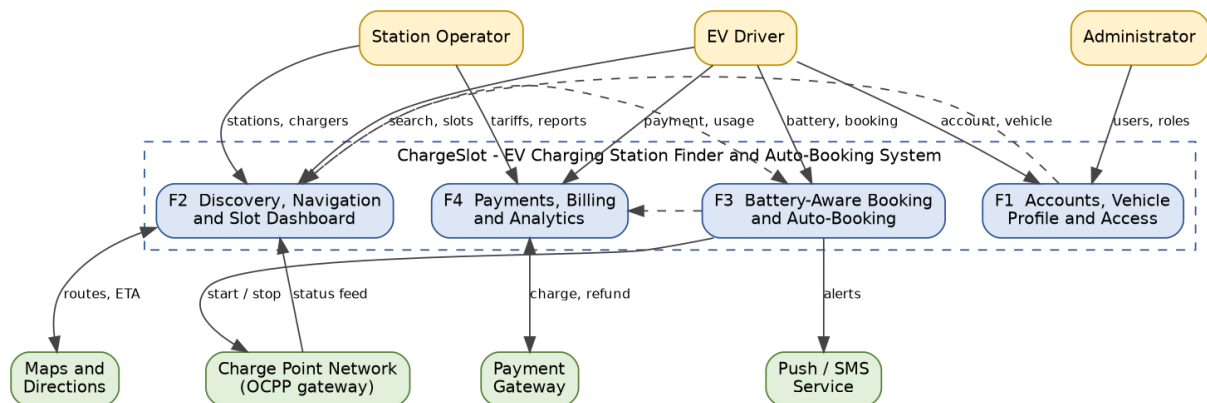


Figure A — System Context Diagram for ChargeSlot

Major interfaces:

- **Charge point gateway** — receives connector status and meter data from charge points over OCPP and sends start and stop commands. In Phase 1 this is backed by a charge point simulator.
- **Vehicle profile** — the battery capacity, connector type and charging rates of the driver's vehicle, produced by the accounts subsystem and consumed by the booking subsystem.
- **Slot inventory** — the 15-minute slots of every connector, maintained by the discovery subsystem and consumed by the booking subsystem.
- **Maps and directions service** — provides map display, routes and travel time.
- **Payment gateway** — collects payments and issues refunds. The system never handles card data.
- **Notification service** — delivers push notifications to the driver's phone.

2.2 Product Functions

For EV drivers:

- Register, log in, and maintain a profile and up to five vehicles.
- Find nearby stations on a map or list, with filters for connector type, power, price and availability.

- See a live slot dashboard showing the status of every connector and the free time slots.
- Get a route preview and travel time, and start navigation to the station.
- Enter the current and target battery level and see the usable range, reachable stations, energy needed, charging time and cost.
- Book a slot manually, or let the system auto-book the best slot.
- Modify or cancel a booking, and be rebooked automatically if a charger fails.
- Check in, start, monitor and stop a charging session from the app.
- Pay in the app, receive invoices, and see usage analytics.

For station operators:

- Add and manage stations, chargers and connectors, and mark chargers out of service.
- Set tariffs, reservation fees, and cancellation and no-show rules.
- View utilisation, revenue, peak hours and charger uptime.

For administrators:

- Manage users and roles, and approve or reject operators and stations.
- Set platform limits, such as maximum tariffs and scoring weights.
- View platform analytics and payment reconciliation results.

Grouping of functions:

Group	Functions
User accounts, vehicle profile and access	Registration, login, roles, driver profile, vehicle profile, operator onboarding, audit
Station discovery, navigation and slot dashboard	Station and charger management, search and filters, live status, slot timeline, route preview, navigation hand-off
Battery-aware booking and auto-booking	Battery input, range, energy and time estimates, manual booking, auto-booking, lifecycle, charging session, notifications
Payments, billing and analytics	Payment methods, reservation fee, final bill, refunds, invoices, tariffs, driver, operator and platform analytics

2.3 User Classes and Characteristics

The system has three human user classes and one automated internal process.

- **EV Driver (primary user).** Uses the system most frequently, often outdoors, one-handed, and in a hurry. Searches for stations, books slots, charges, and pays. Assumed to be comfortable using a smartphone but to have no technical knowledge of charging. This is the most important class to satisfy.
- **Station Operator (secondary user).** Owns or manages one or more stations. Uses the system occasionally to maintain stations, chargers and tariffs, and to read reports. Assumed to have moderate technical familiarity. Can act only on their own stations.
- **Administrator (secondary user).** Approves operators and stations, manages users and roles, and sets platform limits. Assumed to have moderate to high technical familiarity.
- **Booking and slot engine (automated internal process, not a human user).** Generates slots, holds and releases slots, expires unpaid holds, marks no-shows, performs auto-rebooking, and refreshes analytics. It is triggered by time and by system events rather than by a human user.

2.4 Operating Environment

- **Mobile app:** React Native (TypeScript) application for Android 10 or later and iOS 15 or later, with a responsive layout for phone screen sizes.
- **Backend:** Python with the FastAPI framework, providing a REST API and WebSocket channel that handle authentication, station data, slot management, booking logic, billing and analytics.
- **Database:** PostgreSQL with the PostGIS extension for location queries, storing users, vehicles, stations, chargers, slots, bookings, sessions, payments and analytics aggregates. An in-memory cache (Redis) holds slot holds and live connector status.
- **Authentication:** Token-based sessions using JSON Web Tokens (JWT); passwords stored using Argon2 or bcrypt.
- **External dependencies:** Google Maps Platform (maps and directions); a payment gateway in test mode (Razorpay); Firebase Cloud Messaging (push notifications); and a charge point gateway speaking OCPP 1.6J, backed in Phase 1 by a charge point simulator.
- **Hosting:** Deployable on any standard cloud virtual machine or platform-as-a-service that supports Python and containers. The exact provider is a team infrastructure decision.
- **Client requirement:** Internet connectivity and location services are required. If location is denied, the driver may choose a location manually.

2.5 Design and Implementation Constraints

- The technology stack is fixed by team decision: React Native for the mobile app, Python with FastAPI for the backend, and PostgreSQL for persistent storage. All modules must be implemented within this stack for consistency across team members' features.
- Booking and auto-booking logic in Phase 1 is a rule-based, weighted-scoring method, not a trained machine-learning model. It must be designed so that a learned model can replace it later without redesigning other subsystems.
- No real charging hardware is available in Phase 1. Connector status and meter data shall come from a charge point simulator that speaks the same protocol (OCPP 1.6J) as real chargers, and the stations used for testing shall be seeded data.
- The Maps and Directions services are billed and rate-limited. Route and ETA results must be cached and requested only when a driver needs them, not on every screen refresh.
- Payments shall use the gateway's hosted checkout and tokens so that card data never reaches the system's servers. Only the gateway's test mode is used in Phase 1.
- The system must comply with the course-mandated engineering process: version control on GitHub, pull-request-based code review, containerisation with Docker, continuous integration and delivery with Jenkins, unit testing with pytest, and static analysis with SonarQube.
- Personal data, including location, shall be handled in line with the Digital Personal Data Protection Act, 2023, and role-based access shall be enforced on the server.

2.6 Assumptions and Dependencies

- Assumes drivers enter their battery level and vehicle specifications reasonably accurately. The accuracy of range, time and cost estimates depends directly on this.
- Assumes station operators keep their charger details, opening hours and tariffs current.
- Assumes chargers send status updates at least once every 60 seconds, so that the 120-second stale-status rule in Section 5.2 is meaningful.
- Depends on the continued availability, pricing and quota of the Maps, payment and notification services. If one is unavailable the system degrades as described in Section 6.2.

- Assumes the database schema for users, vehicles, stations, slots, bookings and payments is agreed jointly across all four feature owners, since several features read from and write to shared entities.
- Assumes all stations are in a single country, so that one currency (INR), one time zone (IST) and metric units apply in version 1.0.

3. External Interface Requirements

3.1 User Interfaces

Key screens relevant to the overall product:

- **Login / Sign-up screen** — email and password form with inline validation messages and a one-time code step.
- **Home map screen** — map with colour-coded station markers, a search bar, a filter bar and a list toggle.
- **Station detail and slot dashboard** — connectors with live status, a slot timeline for the chosen date, price per kWh, rating and amenities.
- **Battery and booking screen** — current and target battery sliders, usable range, estimated energy, time and cost, recommended slots, and an Auto-Book button.
- **Navigation screen** — route preview with distance and travel time, and a button to open the device's navigation application.
- **My bookings and active session screen** — upcoming bookings with QR code, check-in, live power, energy, battery estimate and cost, and Stop.
- **Payments and invoices screen** — saved methods, payment history, and invoice download.
- **Usage analytics screen** — energy, spend and session charts for the driver.
- **Operator and administrator sections** — role-gated screens for stations, chargers, tariffs, approvals, users and analytics, within the same application.

UI conventions to be followed consistently across all screens:

- A persistent bottom navigation bar with Map, Bookings, Charging, Analytics and Profile.
- Connector status shall use colour and a text label together (Available, Occupied, Reserved, Faulted, Offline) so that it is readable without colour vision.
- Plain-language error messages rather than raw error codes, since the primary user is non-technical.
- Touch targets of at least 44 by 44 points, because the app is often used one-handed at a charger.
- Loading indicators for any screen that depends on a live or external call, and a visible “last updated” time on live data.

Detailed UI mockups and wireframes are documented separately in a design file and are out of scope for this SRS.

3.2 Software Interfaces

Software Interface	Purpose	Input	Output
Mobile app ↔ Backend	Carries user actions and data requests	Login, searches, bookings, session commands, payment requests	Stations, slots, estimates, bookings, session data, invoices, analytics
Backend ↔ Database	Stores and retrieves persistent data	User, vehicle, station, slot, booking, payment records	Stored records and query results
Backend ↔ Charge point gateway	Receives status and meter data, sends start and stop	Status and meter messages; remote start and stop commands	Connector status, energy delivered, command acknowledgement
Backend ↔ Maps and Directions	Provides routes and travel time	Origin, destination, departure time	Route, distance, travel time
Backend ↔ Payment gateway	Collects payments and refunds	Amount, order details, payment token, refund request	Payment result, gateway reference, webhook events
Backend ↔ Notification service	Delivers push notifications	Device token, message	Delivery status
Booking engine ↔ Slot dashboard	Keeps slot availability consistent	Hold, confirm, release events	Updated slot states
Booking engine ↔ Billing	Triggers charges and refunds	Booking and session events	Reservation fee, final bill, refund

3.2.1 Mobile App–Backend Interface

The mobile app shall communicate with the backend through REST APIs and a WebSocket channel. The app shall submit account, vehicle, search, battery, booking, session and payment requests. The backend shall return stations, slot timelines, estimates, bookings, live session data, invoices and analytics, and shall push live connector status changes over the WebSocket channel.

3.2.2 Database Interface

The backend shall communicate with the PostgreSQL database to store and retrieve users, vehicles, stations, chargers, connectors, slots, tariffs, bookings, sessions, payments, invoices, notifications, audit records and analytics aggregates. Location queries shall use the geospatial capability of the database.

3.2.3 Charge Point Gateway Interface

The gateway shall deliver connector status changes and meter values to the backend. The backend shall send remote start and remote stop commands for a connector and shall treat a missing acknowledgement within 15 seconds as a failed command. The backend shall treat the gateway as the source of truth for connector status.

3.2.4 Maps and Directions Interface

The backend shall request routes and travel times from the maps service for a given origin and destination and shall cache identical requests for up to 2 minutes. The app shall render the map and route preview and shall hand off turn-by-turn guidance to the device's navigation application by deep link.

3.2.5 Payment Gateway Interface

The backend shall create payment orders with the gateway and verify each result with the gateway from the server. The gateway shall notify the backend of payment events by signed webhooks. The backend shall use idempotency keys on every charge and refund request.

3.2.6 Notification Interface

The backend shall send booking, reminder, session and payment notifications to the notification service for delivery to the driver's device. A failure in the notification service shall not block any other operation.

3.3 Communications Interfaces

- Client-server communication shall use HTTPS with TLS 1.2 or later, and live updates shall use secure WebSocket (wss).
- Backend services shall expose REST APIs. Request and response bodies shall use JSON.
- Timestamps shall be exchanged in ISO 8601 format in UTC and displayed in the time zone of the device.
- Authentication tokens shall be sent in the request header and never in a URL.
- Payment webhooks shall be accepted only over HTTPS and only with a valid signature.
- Standard HTTP status codes shall be used for success and error conditions.

3.4 Hardware Interfaces

The system requires no dedicated hardware of its own. Drivers use a smartphone with GPS and, for check-in by QR code, a camera. Station operators and administrators use the same application on a smartphone or tablet. Charging hardware belongs to the station operators and is reached only through the charge point gateway using the OCPP protocol.

4. Analysis Models

4.1 Use Case Diagram

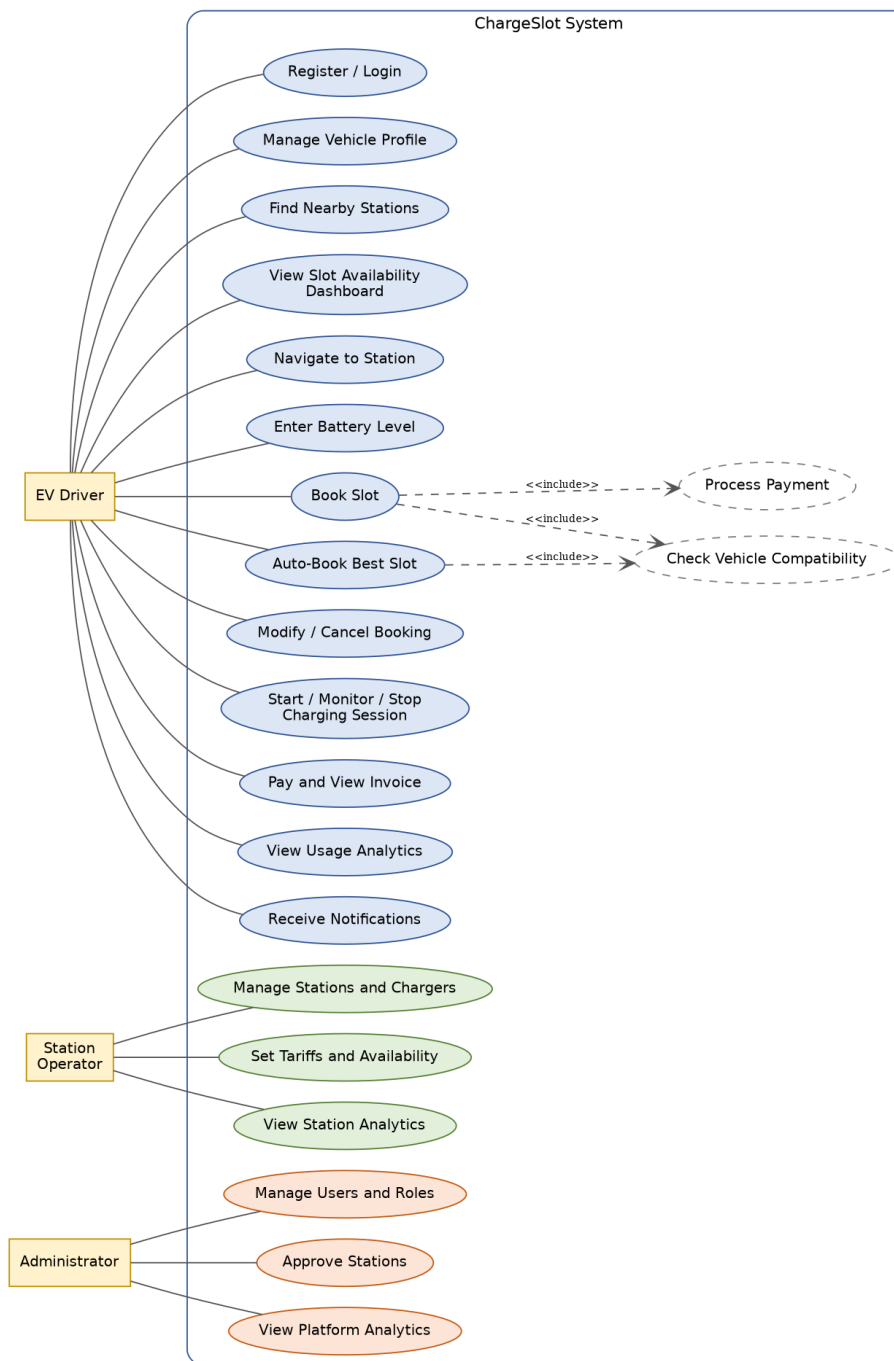


Figure 1 — Use Case Diagram for ChargeSlot

The system has three actors: EV Driver, Station Operator, and Administrator.

- **Driver use cases:** Register / Login, Manage Vehicle Profile, Find Nearby Stations, View Slot Availability Dashboard, Navigate to Station, Enter Battery Level, Book Slot, Auto-Book Best Slot, Modify / Cancel Booking, Start / Monitor / Stop Charging Session, Pay and View Invoice, View Usage Analytics, and Receive Notifications.

- **Station Operator use cases:** Manage Stations and Chargers, Set Tariffs and Availability, and View Station Analytics.
- **Administrator use cases:** Manage Users and Roles, Approve Stations, and View Platform Analytics.

Book Slot and Auto-Book Best Slot both include Check Vehicle Compatibility, and Book Slot includes Process Payment. The booking workflow shows that battery level and vehicle data decide the compatible connectors and the slot length, and that the confirmed booking drives the charging session and the bill.

4.2 Entity Relationship Diagram

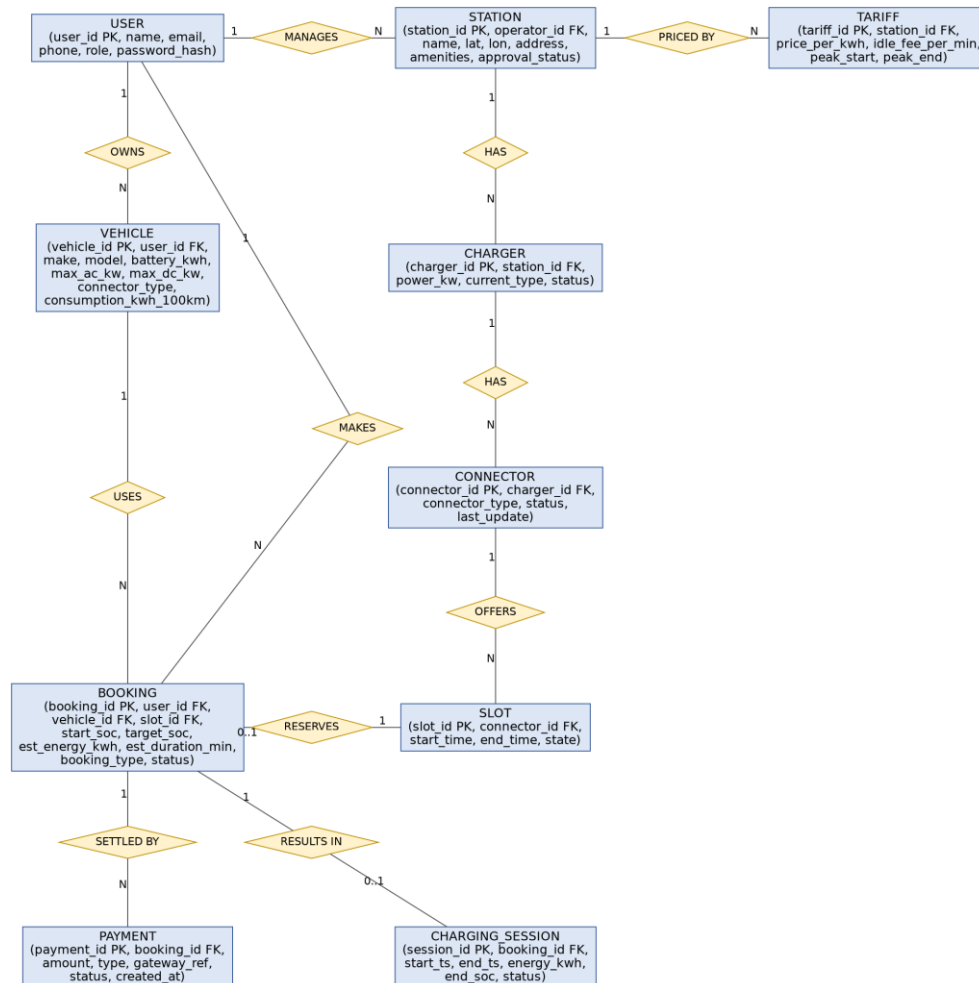


Figure 2 — Entity Relationship Diagram for ChargeSlot

The core domain entities and their relationships are:

- **USER** (user_id, name, email, phone, role, password_hash) — OWNS many VEHICLES; MAKES many BOOKINGS; a user with the Station Operator role MANAGES many STATIONS.
- **VEHICLE** (vehicle_id, user_id, make, model, battery_kwh, max_ac_kw, max_dc_kw, connector_type, consumption_kwh_100km) — USES: one vehicle is used in many BOOKINGS.
- **STATION** (station_id, operator_id, name, lat, lon, address, amenities, approval_status) — HAS many CHARGERS; is PRICED BY many TARIFFS.
- **CHARGER** (charger_id, station_id, power_kw, current_type, status) — HAS many CONNECTORS.

- **CONNECTOR** (connector_id, charger_id, connector_type, status, last_update) — OFFERS many SLOTS.
- **SLOT** (slot_id, connector_id, start_time, end_time, state) — RESERVES: a slot is reserved by zero or one BOOKING.
- **BOOKING** (booking_id, user_id, vehicle_id, slot_id, start_soc, target_soc, est_energy_kwh, est_duration_min, booking_type, status) — RESULTS IN zero or one CHARGING_SESSION; is SETTLED BY many PAYMENTS.
- **CHARGING_SESSION** (session_id, booking_id, start_ts, end_ts, energy_kwh, end_soc, status).
- **PAYMENT** (payment_id, booking_id, amount, type, gateway_ref, status, created_at).
- **TARIFF** (tariff_id, station_id, price_per_kwh, idle_fee_per_min, peak_start, peak_end).

Notification, audit, favourite and review entities are maintained by the corresponding subsystems and are not drawn in Figure 2. Field lengths and types are given in Appendix B.

4.3 Booking Lifecycle State Diagram

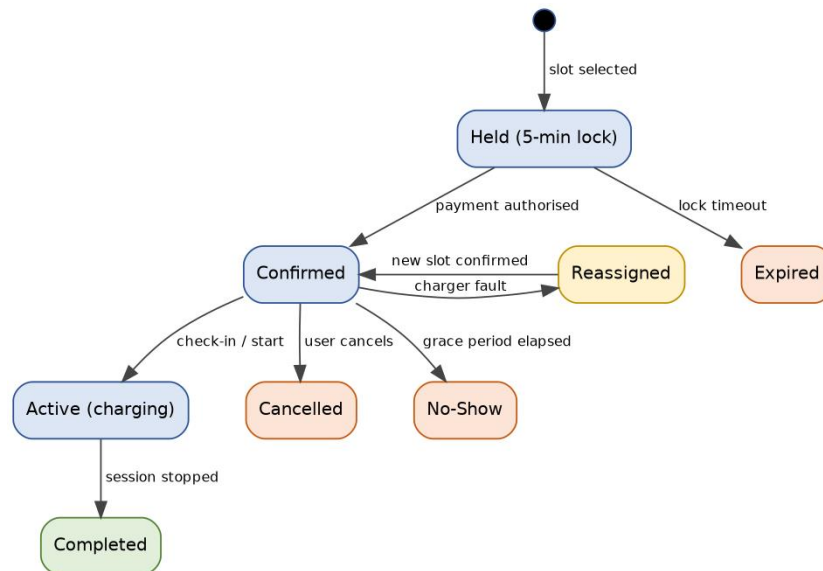


Figure 3 — Booking Lifecycle State Diagram

A booking starts as Held while the driver pays, becomes Confirmed after the gateway confirms payment, and becomes Active when charging starts. It ends as Completed, Cancelled, No-Show or Expired. If a charger fails while the booking is Confirmed, the booking becomes Reassigned until a replacement slot is confirmed. Only the transitions drawn in Figure 3 are permitted (REQ-F3-22).

5. System Features

5.1 User Accounts, Vehicle Profile and Access Control

5.1.1 Description and Priority

This feature creates and protects the identity of every user and holds the vehicle data on which the rest of the system depends. A driver registers, verifies their contact details, and saves one or more vehicles with battery capacity, connector type and charging rates. The system assigns every user one of three roles and enforces what that role may do on the server. Station operators are onboarded through an approval step.

This feature is the entry point to the system. The booking feature reads the vehicle profile to decide which connectors are compatible and how long a charge takes, and every feature relies on role-based access. If vehicle data is wrong, every estimate that follows from it is wrong.

Priority: High.

Rating	Score (1 to 9)	Reason
Benefit	8	No booking, payment or analytics is possible without an authenticated user and a vehicle
Penalty	9	If absent, there is no secure access and no way to know which connectors suit a vehicle
Cost	4	Standard authentication and CRUD, with role enforcement on every endpoint
Risk	3	Well-understood patterns; the main risk is inconsistent access checks across modules

5.1.2 Stimulus/Response Sequences

Sequence 1: Driver registers and adds a vehicle

Step	Actor	Action or response
1	Driver	Opens the app and chooses Sign up, entering email, phone and password
2	System	Validates the fields and password rule, creates the account with the Driver role, and sends a one-time code
3	Driver	Enters the code
4	System	Marks the contact as verified and opens the profile and vehicle screen
5	Driver	Picks a preset EV model or enters battery capacity, connector type, charging rates and consumption
6	System	Validates the values against the allowed ranges, saves the vehicle, and sets it as default if it is the first one

Sequence 2: Repeated failed login

Step	Actor	Action or response
1	User	Enters an incorrect password
2	System	Shows "Invalid email or password" and counts the failure
3	User	Fails 5 times in a row
4	System	Locks the account for 15 minutes and notifies the account holder
5	User	Tries again within 15 minutes
6	System	Refuses the attempt and shows when the account will unlock

Sequence 3: Operator onboarding

Step	Actor	Action or response
1	User	Applies for Station Operator status with a business name and contact details
2	System	Stores the application as pending and notifies Administrators
3	Administrator	Reviews the application and approves or rejects it with a reason
4	System	Records the decision in the audit log, changes the role on approval, and notifies the applicant

5.1.3 Functional Requirements

Registration and authentication

REQ-F1-01: The system shall allow a new user to register using a unique email address, a mobile phone number, and a password.

REQ-F1-02: The system shall require a password of at least 8 characters containing at least one letter and one digit, and shall reject a weaker password with a message that states the rule.

REQ-F1-03: The system shall verify the user's email address or phone number with a one-time code before the account is allowed to create a booking.

REQ-F1-04: The system shall authenticate a registered user by email and password and shall issue an access token valid for 30 minutes and a refresh token valid for 7 days.

REQ-F1-05: The system shall lock an account for 15 minutes after 5 consecutive failed login attempts and shall notify the account holder.

REQ-F1-06: The system shall allow a user to reset a forgotten password using a single-use code that expires after 15 minutes.

REQ-F1-07: The system shall allow a user to log out, and shall invalidate the refresh token on logout.

REQ-F1-08: The system shall reject a registration whose email address or phone number already belongs to an existing account.

REQ-F1-09: The system shall respond to a failed login with the generic message "Invalid email or password" without revealing which of the two was wrong.

Roles and access control

REQ-F1-10: The system shall assign exactly one role to every account: Driver, Station Operator, or Administrator. Self-registered accounts shall default to Driver.

REQ-F1-11: The system shall enforce role-based access on the server for every protected endpoint and shall return an authorization error when the caller's role is insufficient.

REQ-F1-12: The system shall restrict a Station Operator to the stations, chargers, tariffs and reports that belong to that operator.

REQ-F1-13: The system shall allow an Administrator to create, deactivate and reactivate user accounts and to change a user's role.

REQ-F1-14: The system shall record an audit entry, naming the acting user, for every role change, account deactivation, and operator approval or rejection.

REQ-F1-15: The system shall allow a user to apply for Station Operator status by submitting a business name and contact details, and shall allow an Administrator to approve or reject the application with a recorded reason.

Driver profile and preferences

REQ-F1-16: The system shall allow a driver to view and edit their name, phone number and preferences. Preferences are: default charging mode (Fastest, Cheapest or Nearest), default target battery level, default search radius, and notification settings.

REQ-F1-17: The system shall reject a profile save in which a required field is empty or a phone number is malformed, and shall identify each invalid field inline.

REQ-F1-18: The system shall ask for location permission and, if it is denied, shall continue to work with a manually chosen location (typed place name or map pin).

REQ-F1-19: The system shall not retain a driver's raw location coordinates for more than 24 hours after they were received.

REQ-F1-20: The system shall allow a driver to request deletion of their account. The request shall be refused while an upcoming booking or an unpaid balance exists, and the reason shall be displayed.

Vehicle profile

REQ-F1-21: The system shall allow a driver to add, edit and delete vehicles. Each vehicle shall store make, model, battery capacity (kWh), connector type, maximum AC charging rate (kW), maximum DC charging rate (kW), energy consumption (kWh per 100 km) and an optional registration number.

REQ-F1-22: The system shall allow at most 5 vehicles per driver and exactly one default vehicle.

REQ-F1-23: The system shall offer a preset list of common EV models that pre-fills the specification fields, which the driver may then edit.

REQ-F1-24: The system shall reject vehicle values outside the ranges defined in Appendix B.

REQ-F1-25: The system shall prevent deletion of a vehicle that is linked to an upcoming or active booking.

REQ-F1-26: The system shall ensure that a driver can view and modify only their own profile and vehicles.

5.2 Station Discovery, Navigation and Real-Time Slot Dashboard

5.2.1 Description and Priority

This feature lets operators describe their stations and lets drivers find them. It keeps a live dashboard of the status of every connector and of the free and booked time slots, and it helps the driver reach the station with a route preview and a hand-off to navigation. Live status arrives from the charge point gateway and is pushed to every driver who has the station open.

The demanding part of this feature is trust in live data: a driver who is told a charger is free and finds it occupied loses confidence in the whole product. The system therefore shows when each status was last updated and shows "Status unknown" rather than an old value.

Priority: High. This is the core of what the driver sees and a prerequisite for booking (5.3).

Rating	Score (1 to 9)	Reason
Benefit	9	Finding a working charger and seeing live availability is the main reason a driver uses the product
Penalty	8	Without it, a driver cannot choose a station or a slot
Cost	7	Real-time ingestion, slot generation, geospatial search and maps integration
Risk	7	Accuracy and delay of live data, and dependence on the gateway and maps service

5.2.2 Stimulus/Response Sequences

ID	Stimulus	System Response
SR-F2-01	Driver opens the home map	System shows nearby stations within the radius with availability markers
SR-F2-02	Driver applies filters or sorts	System returns matching stations in the chosen order, paginated
SR-F2-03	Driver opens a station	System shows connectors with live status and time of last update, the price and the average rating
SR-F2-04	Charger reports a status change	System updates the connector and pushes the change to every open dashboard of that station
SR-F2-05	No update received from a connector for more than 120 seconds	System shows "Status unknown" for that connector
SR-F2-06	Driver selects a date on the slot timeline	System shows each connector's 15-minute slots as Free, Booked, Blocked or Past
SR-F2-07	Driver taps Navigate	System shows the route preview and, on request, opens the navigation application
SR-F2-08	Driver with a confirmed booking is running late	System warns the driver and offers to reschedule, extend or cancel
SR-F2-09	Maps service is unavailable	System shows straight-line distance with a notice and continues to work
SR-F2-10	Operator marks a charger out of service	System removes the affected slots from availability and triggers rebooking of affected bookings

5.2.3 Functional Requirements

Station and charger management

REQ-F2-01: The system shall allow a Station Operator to create, edit and deactivate stations. Each station shall store name, address, coordinates, amenities, operating hours and contact details.

REQ-F2-02: The system shall allow an operator to add, edit and remove the chargers and connectors of their station, each with a power rating, current type (AC or DC), connector type, and the identifier of the matching charge point in the gateway.

REQ-F2-03: The system shall keep a new or edited station hidden from drivers until an Administrator approves it.

REQ-F2-04: The system shall validate coordinates, required fields and connector types before saving a station or charger, and shall reject invalid data with a field-level message.

REQ-F2-05: The system shall allow an operator to mark a charger out of service with a reason and an expected return time.

REQ-F2-06: The system shall allow an operator to define maintenance windows, and shall remove the affected slots from availability for those windows.

Station discovery

REQ-F2-07: The system shall display nearby stations on a map and in a list, based on the driver's current or chosen location, within a selectable radius (default 10 km, range 1 to 50 km).

REQ-F2-08: The system shall colour-code each station marker by availability and shall show the number of available connectors out of the total, using a text label as well as colour.

REQ-F2-09: The system shall allow stations to be filtered by connector type, minimum power, availability now, maximum price per kWh, amenities, and compatibility with the selected vehicle.

REQ-F2-10: The system shall allow results to be sorted by distance, price, power, or availability.

REQ-F2-11: The system shall allow stations to be searched by name, address or area.

REQ-F2-12: The system shall paginate list results (20 per page) and shall cluster map markers when more than 50 stations are in view.

REQ-F2-13: The system shall display a station detail screen with name, address, operator, amenities, operating hours, each connector's type, power and live status, the price per kWh, and the average rating.

REQ-F2-14: The system shall allow a driver to save and remove favourite stations.

REQ-F2-15: The system shall allow a driver to rate (1 to 5) and review a station once per completed session.

REQ-F2-16: The system shall allow a driver to report a faulty charger and shall forward the report to the station's operator.

Real-time slot dashboard

REQ-F2-17: The system shall ingest status updates from the charge point gateway and shall record, for each connector, one of the statuses Available, Occupied, Reserved, Faulted or Offline together with the time of the update.

REQ-F2-18: The system shall display each connector's current status and the time it was last updated.

REQ-F2-19: The system shall deliver a status change to every app that has the affected station's dashboard open within 10 seconds for at least 95% of changes.

REQ-F2-20: The system shall display "Status unknown" for a connector from which no update has been received for more than 120 seconds, instead of showing the last known status.

REQ-F2-21: The system shall display a Reserved status for a connector that has a confirmed booking starting within the next 15 minutes.

REQ-F2-22: The system shall generate bookable slots of 15 minutes for every connector for up to 7 days ahead, within operating hours and excluding maintenance windows.

REQ-F2-23: The system shall display, for a selected station and date, a slot timeline per connector in which each slot is shown as Free, Booked, Blocked or Past.

REQ-F2-24: The system shall display an estimated wait time when all compatible connectors at a station are occupied, calculated from the expected end of the active sessions.

Navigation

REQ-F2-25: The system shall display a route preview from the driver's location to a station, showing the route, road distance and estimated travel time with current traffic.

REQ-F2-26: The system shall allow the driver to start turn-by-turn navigation in the device's navigation application through a deep link.

REQ-F2-27: The system shall, for a confirmed booking, compare the driver's estimated arrival time with the slot start time, and shall warn the driver and offer to reschedule, extend or cancel when the projected arrival is later than the slot start plus the grace period.

REQ-F2-28: The system shall cache route results for identical origin and destination pairs for up to 2 minutes, and shall show straight-line distance with a notice when the maps service is unavailable.

5.2.4 Feature Constraints and Validation

- A station is not visible to drivers until it has been approved by an Administrator.
- Connector types shall be selected from the supported list in Appendix B.
- Slots are generated only within opening hours and never overlap maintenance windows.
- A connector status shall never be shown as Available when a confirmed booking starts within the next 15 minutes.

5.3 Battery-Aware Booking and Auto-Booking

5.3.1 Description and Priority

This feature turns the driver's battery level into a booking. From the current and target battery level, the vehicle profile and the live slot inventory, the system works out how far the car can still go, which stations it can reach, how much energy it needs and how long that takes on each compatible connector. It then recommends slots, or books the best one automatically using a weighted score. It also manages the booking from payment hold to completion, handles cancellation, no-shows and charger faults, and runs the charging session.

The demanding parts are preventing two drivers from booking the same slot at the same moment, and keeping estimates honest. Estimates are presented as guidance, not as a guarantee.

Priority: High. This is the core value proposition of the product.

Rating	Score (1 to 9)	Reason
Benefit	9	Matching the booking to the battery is what distinguishes the product from a plain station finder
Penalty	9	If absent, there is no booking, no auto-booking and no charging session
Cost	8	Estimation, scoring, concurrency control, lifecycle rules and gateway commands
Risk	8	Double-booking under concurrent requests and the accuracy of charging time estimates

5.3.2 Stimulus/Response Sequences

Sequence 1: Driver books a slot with a battery level

Step	Actor	Action or response
1	Driver	Opens a station and enters current battery 25% and target 80%
2	System	Checks the vehicle's connector type and shows only compatible connectors
3	System	Computes energy needed, charging time per connector, slot length and a cost estimate
4	System	Shows up to 5 recommended slots, ranked by the driver's charging mode
5	Driver	Chooses a slot
6	System	Holds the slot for 5 minutes and opens payment of the reservation fee
7	Driver	Pays
8	System	Confirms the payment with the gateway, confirms the booking, and shows the booking reference and QR code

Sequence 2: Auto-booking

Step	Actor	Action or response
1	Driver	Taps Auto-Book, with battery level, target level, start time and charging mode
2	System	Collects compatible, reachable candidate slots across nearby stations
3	System	Scores the candidates with the weights of the charging mode and selects the best
4	System	Shows the chosen slot and the reasons for the choice (Confirm first mode), or books it directly (Book immediately mode)
5	Driver	Confirms
6	System	Holds the slot, takes the reservation fee, and confirms the booking
7	System	If no suitable slot exists, shows the closest alternatives instead

Sequence 3: Cancellation and no-show

Step	Actor	Action or response
1	Driver	Cancels 45 minutes before the start
2	System	Cancels without a fee, frees the slot, and refunds the reservation fee
3	Driver	(Different booking) does not arrive and does not check in
4	System	After the 15-minute grace period marks the booking No-Show, releases the slot, and applies the no-show fee

Sequence 4: Charger fault and auto-rebooking

Step	Actor	Action or response
1	Gateway	Reports that a booked connector is Faulted
2	System	Sets the booking to Reassigned and looks for a slot at the same station within 60 minutes
3	System	If none is found, looks at the nearest compatible reachable station
4	System	Confirms the replacement at no extra cost and notifies the driver
5	System	If no replacement exists, cancels the booking, refunds in full, and notifies the driver

Sequence 5: Charging session

Step	Actor	Action or response
1	Driver	Arrives and scans the QR code at the charger
2	System	Checks the booking and connector status, and sends the remote start command
3	System	Shows live power, energy, battery estimate and cost every 30 seconds
4	System	Stops the session when the energy for the target level has been delivered, or when the driver taps Stop
5	System	Sets the booking to Completed and passes the energy delivered to billing

5.3.3 Functional Requirements

Battery input and estimation

REQ-F3-01: The system shall allow a driver to enter the current battery level (0 to 100%) and a target battery level (default 80%) for the selected vehicle.

REQ-F3-02: The system shall reject a target level that is not at least 5 percentage points above the current level, and shall state the allowed range.

REQ-F3-03: The system shall compute the usable driving range from the battery level, battery capacity, vehicle consumption, and the reserve defined in Appendix B.

REQ-F3-04: The system shall mark each station as reachable or not reachable using the usable range and the road distance, shall show the estimated battery level on arrival, and shall not offer an unreachable station for booking or auto-booking.

REQ-F3-05: The system shall offer only connectors that match the selected vehicle's connector type and current type.

REQ-F3-06: The system shall compute the energy required (kWh) to move from the current to the target level using the formula in Appendix B.

REQ-F3-07: The system shall compute the estimated charging time for each compatible connector using the lower of the connector power and the vehicle's maximum rate for that current type, including the reduced rate above 80% on DC charging.

REQ-F3-08: The system shall set the booked slot length to the estimated charging time plus a 10% margin, rounded up to the next 15 minutes, with a minimum of 15 and a maximum of 240 minutes.

REQ-F3-09: The system shall display, before confirmation, a cost estimate that itemises energy cost, reservation fee and taxes.

Manual booking

REQ-F3-10: The system shall show up to 5 recommended slots for the chosen station, ranked by the selected charging mode.

REQ-F3-11: The system shall allow a driver to choose any free window of the required length on a compatible connector, from 5 minutes ahead up to 7 days ahead.

REQ-F3-12: The system shall hold the chosen slot for 5 minutes while payment is completed and shall release it automatically if payment is not completed.

REQ-F3-13: The system shall never confirm two bookings that overlap on the same connector, including when requests arrive at the same time. The later request shall receive the message "Slot no longer available" with alternative slots.

REQ-F3-14: The system shall reject a booking that overlaps another booking of the same driver or the same vehicle.

REQ-F3-15: The system shall confirm a booking only after the gateway confirms that the reservation fee has been paid, and shall then issue a booking reference and a QR code.

Auto-booking

REQ-F3-16: The system shall provide an Auto-Book action that, given a vehicle, battery level, target level, preferred start time and an optional destination, selects the best compatible and reachable slot without the driver choosing one.

REQ-F3-17: The system shall rank candidate slots by a weighted score of arrival time, price, charging time, detour distance and station reliability, with the weights set by the charging mode (Fastest, Cheapest, Nearest) as defined in Appendix B.

REQ-F3-18: The system shall show the driver why the slot was chosen, naming the deciding factors.

REQ-F3-19: The system shall support two confirmation modes: Confirm first (default) and Book immediately. Book immediately shall be available only when a saved payment method exists.

REQ-F3-20: The system shall, when the driver has enabled low-battery assistance and the battery level entered is at or below the threshold (default 20%), propose or create an auto-booking according to the confirmation mode.

REQ-F3-21: The system shall, when no compatible reachable slot exists, display a plain-language message with the closest alternatives, such as a lower target level, a later time or a wider search radius.

Booking lifecycle

REQ-F3-22: The system shall manage booking status as Held, Confirmed, Active, Completed, Cancelled, No-Show, Expired and Reassigned, shall allow only the transitions shown in Figure 3, and shall reject any other.

REQ-F3-23: The system shall allow a driver to change the time or connector of a confirmed booking up to 30 minutes before its start, subject to availability.

REQ-F3-24: The system shall allow a driver to cancel a booking, free of charge when at least 30 minutes before the start and with the cancellation fee in the booking's tariff after that.

REQ-F3-25: The system shall mark a booking No-Show when the driver has not checked in within the grace period (default 15 minutes) after the start time, shall release the slot, and shall apply the no-show fee.

REQ-F3-26: The system shall, when a charger becomes Faulted or an operator cancels, automatically look for an alternative slot at the same station within 60 minutes of the original time, then at the nearest compatible station, shall notify the driver, and shall cancel with a full refund if none is found.

Charging session

REQ-F3-27: The system shall allow a driver to check in by scanning the QR code at the charger or by tapping Check in within 300 metres of the station.

REQ-F3-28: The system shall start a session by sending a remote start command through the gateway, and only when the booking is valid for the current time and the connector is available.

REQ-F3-29: The system shall display live power (kW), energy delivered (kWh), estimated battery level, elapsed time and cost so far, refreshed at least every 30 seconds.

REQ-F3-30: The system shall stop a session on the driver's command, or automatically when the energy delivered reaches the planned energy for the target level, by sending a remote stop command.

REQ-F3-31: The system shall display an error and shall not bill the driver when the gateway fails to start a session, and shall offer a retry or another connector.

REQ-F3-32: The system shall start an idle fee when a vehicle remains connected for more than 10 minutes after the session ended, and shall remind the driver before the fee starts.

REQ-F3-33: The system shall send notifications for booking confirmation, a reminder 15 minutes before the start, ETA warnings, session start, target level reached, session end, cancellation and rebooking, according to the driver's notification settings.

5.3.4 Feature Constraints and Validation

- Auto-booking shall never select an incompatible connector or an unreachable station.
- Estimates of range, time and cost shall be labelled as estimates.
- A booking shall not be confirmed until payment is confirmed by the gateway from the server.
- A slot hold shall expire after 5 minutes without payment, and the slot shall then be released immediately.

5.4 Payments, Billing and Usage Analytics

5.4.1 Description and Priority

This feature takes payment for bookings and sessions, produces correct bills and invoices, and turns completed sessions into usage analytics. A reservation fee is paid when a booking is confirmed. When the session ends, the system computes the final bill from the energy delivered and the tariff that applied at booking, deducts the reservation fee, and collects the balance. Operators set tariffs and read earnings. Drivers, operators and administrators each see analytics limited to what their role permits.

Payment correctness is critical: the driver must never be charged twice and the system must never confirm a booking that was not paid.

Priority: High for payments and billing; Medium for analytics.

Rating	Score (1 to 9)	Reason
Benefit	8	Payment is required to confirm a booking, and analytics give drivers and operators lasting value
Penalty	8	Without payment there are no confirmed bookings; without analytics the product still works
Cost	7	Gateway integration, refunds, reconciliation, invoices and aggregation
Risk	7	Duplicate or lost payments, webhook handling, and mismatch between records and the gateway

5.4.2 Stimulus/Response Sequences

ID	Stimulus	System Response
SR-F4-01	Driver confirms a slot and chooses a payment method	System creates a payment order for the reservation fee through the gateway
SR-F4-02	Gateway reports a successful payment	System verifies it with the gateway, then confirms the booking
SR-F4-03	Payment fails or is abandoned	System keeps the slot held until the hold expires, then releases it and tells the driver
SR-F4-04	Session ends	System computes the final bill, deducts the reservation fee, and collects the balance
SR-F4-05	Balance payment fails	System marks the account Payment Due and blocks new bookings until it is settled
SR-F4-06	Booking is cancelled or cannot be rebooked	System refunds according to the cancellation policy
SR-F4-07	Session is completed and paid	System generates a PDF invoice and shows it in payment history
SR-F4-08	Operator changes a tariff	System applies it to future bookings only
SR-F4-09	Driver opens usage analytics	System shows energy, spend, sessions and trends for the chosen period
SR-F4-10	Operator or Administrator exports a report	System produces a CSV limited to the data their role permits

5.4.3 Functional Requirements

Payment methods

REQ-F4-01: The system shall allow a driver to pay by UPI, debit or credit card, or net banking through the payment gateway.

REQ-F4-02: The system shall allow a driver to save a payment method as a gateway token, and shall never store card numbers, CVV or UPI PIN.

REQ-F4-03: The system shall allow a driver to remove a saved payment method.

Charges, settlement and refunds

REQ-F4-04: The system shall collect a reservation fee at booking confirmation, set per station (default INR 50), and shall credit it against the final bill.

REQ-F4-05: The system shall compute the final bill as energy delivered multiplied by the tariff, plus idle fee and taxes, minus the reservation fee already paid.

REQ-F4-06: The system shall apply the tariff, including time-of-use periods, that was in force when the booking was confirmed, even if the operator changes the tariff later.

REQ-F4-07: The system shall collect the balance at session end from the saved payment method, or request payment, and shall mark the account Payment Due and block new bookings if the payment fails until it is settled.

REQ-F4-08: The system shall refund cancellations and failed rebookings according to the booking's cancellation policy, to the original payment method.

REQ-F4-09: The system shall record every payment attempt and its outcome (success, failure, pending, refunded) with the gateway reference.

REQ-F4-10: The system shall retry a failed charge at most 3 times and shall use idempotency keys so that a retry can never charge a driver twice.

REQ-F4-11: The system shall process payment gateway webhooks only after verifying their signature, and shall apply each event only once.

REQ-F4-12: The system shall treat a payment as successful only after the server has confirmed it with the gateway, and shall not rely on a success message sent by the app.

REQ-F4-13: The system shall run a daily reconciliation of its payment records against the gateway report and shall flag mismatches to an Administrator.

Invoices and tariffs

REQ-F4-14: The system shall generate a PDF invoice for every completed session showing invoice number, date, station, energy, rate, idle fee, taxes, total and payment method.

REQ-F4-15: The system shall allow a driver to view and download invoices and to view payment history filtered by date.

REQ-F4-16: The system shall allow an operator to set and change tariffs: price per kWh, time-of-use periods, idle fee, reservation fee, and cancellation and no-show fees, within limits set by the Administrator. Changes shall apply to future bookings only.

REQ-F4-17: The system shall show an operator their earnings per period: gross amount, platform commission (configurable) and net amount.

Usage analytics

REQ-F4-18: The system shall show a driver, for a selected period, the total energy charged, number of sessions, total spend, average cost per kWh and average session duration.

REQ-F4-19: The system shall show a driver monthly trends of energy and spend as charts.

REQ-F4-20: The system shall show a driver their most visited stations and the estimated CO2 avoided, using a configurable emission factor.

REQ-F4-21: The system shall show an operator, per station and per charger, utilisation, peak hours, revenue, energy delivered, average session length, cancellation and no-show rates, and charger uptime.

REQ-F4-22: The system shall show an Administrator platform totals: active users, bookings per day, revenue, payment failure rate and the busiest stations.

REQ-F4-23: The system shall allow analytics to be filtered by the last 7, 30 or 90 days or a custom date range.

REQ-F4-24: The system shall allow operators and Administrators to export reports as CSV.

REQ-F4-25: The system shall limit analytics to the data the user's role may see: a driver sees only their own data, an operator sees only their own stations without personal details of drivers, and an Administrator sees aggregated data.

REQ-F4-26: The system shall base analytics on completed sessions and shall refresh them within 5 minutes of a session completing, showing the time of the last refresh.

5.4.4 Feature Constraints and Validation

- Card numbers, CVV and UPI PIN shall never reach or be stored by the system.
- A payment shall be recorded once only, even if the gateway sends the same event more than once.
- Amounts shall be stored in the smallest currency unit (paise) to avoid rounding errors.
- Tariff values shall stay within the limits set by the Administrator.
- Analytics shall never expose another driver's personal details to an operator.

6. Other Nonfunctional Requirements

6.1 Performance Requirements

ID	Requirement
NFR-PERF-01	Nearby-station searches should return results within 2 seconds for at least 95% of requests under the expected project workload.
NFR-PERF-02	Authenticated API requests should normally receive a response within 2 seconds, excluding calls to external services.
NFR-PERF-03	Slot recommendation and auto-booking requests should complete within 5 seconds for a typical vehicle and station area.
NFR-PERF-04	A live connector status change shall reach an open dashboard within 10 seconds for at least 95% of changes.
NFR-PERF-05	Booking confirmation, once payment is verified, should complete within 3 seconds.
NFR-PERF-06	The app should show its first screen within 3 seconds of launch on a mid-range phone.
NFR-PERF-07	The system shall paginate large result sets rather than returning an unbounded collection.
NFR-PERF-08	The system should support at least 500 concurrent users without noticeable degradation of search, booking and dashboard functions.

6.2 Safety Requirements

ID	Requirement
NFR-SAFE-01	The system shall present range, charging time and cost as estimates, and shall not present them as guarantees.
NFR-SAFE-02	The system shall not start a charging session on a connector that is incompatible with the booked vehicle or that is reported Faulted.
NFR-SAFE-03	A stop command from the driver shall always be sent to the gateway, and the app shall tell the driver if it was not acknowledged.
NFR-SAFE-04	The system shall not require a driver to operate the app while driving; navigation shall be handled by the device's navigation application.
NFR-SAFE-05	The system shall fail safely when an external service (maps, payment, gateway, notifications) is unavailable: it shall show a clear message, shall not lose or double-charge a payment, and shall not start a session it cannot control.
NFR-SAFE-06	The system shall handle invalid, missing or unexpectedly formatted data from the gateway without crashing.

6.3 Security Requirements

ID	Requirement
NFR-SEC-01	The system shall require authentication before access to any protected data or operation.
NFR-SEC-02	The system shall enforce role-based authorization on the server for every protected operation.
NFR-SEC-03	Passwords shall never be stored in plaintext and shall use Argon2 or bcrypt hashing.
NFR-SEC-04	Access tokens shall expire within 30 minutes and shall be stored in the device's secure storage.
NFR-SEC-05	All traffic between the app and the backend shall use TLS 1.2 or later.
NFR-SEC-06	The system shall validate and sanitise all input to reduce the risk of injection and malformed requests, and shall rate-limit login, one-time code and booking requests.
NFR-SEC-07	Payment card data shall be handled only by the payment gateway; the system's scope shall be limited to gateway tokens.
NFR-SEC-08	A driver shall be able to access only their own profile, vehicles, bookings, sessions, payments and analytics.
NFR-SEC-09	An operator shall be able to access only data for their own stations.
NFR-SEC-10	Administrative actions, tariff changes, role changes and payment adjustments shall be auditable and attributable to an authenticated user.
NFR-SEC-11	Application errors shall not expose passwords, tokens, stack traces or other confidential details to end users.

6.4 Software Quality Attributes

ID	Requirement
NFR-QUAL-01	Usability — A driver should be able to find a station and book a slot in no more than 5 screen steps from the home map.
NFR-QUAL-02	Maintainability — Major modules (accounts, discovery, booking, payments, analytics) shall remain logically separated.
NFR-QUAL-03	Testability — Estimation, scoring, state transitions, authorization, billing and refund logic shall be implementable as isolated automated tests.
NFR-QUAL-04	Reliability — A failure in a non-critical service such as notifications shall not prevent search, booking or payment from working.
NFR-QUAL-05	Availability — The backend should be available at least 99% of the time during the evaluation period.
NFR-QUAL-06	Scalability — The data model shall support growth in stations, connectors and users without redesign of core entities.
NFR-QUAL-07	Portability — The app shall run on Android 10 or later and iOS 15 or later.
NFR-QUAL-08	Accessibility — Screens should provide readable text, labels for controls, sufficient contrast, and status shown by text as well as colour.
NFR-QUAL-09	Interoperability — The system shall communicate with chargers using OCPP 1.6J and with external services through their published APIs.
NFR-QUAL-10	Consistency — Connector types, status values and booking states shall use the same identifiers across screens, APIs and reports.

6.5 Business Rules

ID	Requirement
BR-01	Only approved and active stations, chargers and connectors are visible to drivers and available for booking.
BR-02	A booking may only be made on a connector that matches the vehicle's connector type and current type.
BR-03	A connector shall never have two confirmed bookings that overlap in time.
BR-04	A booking is confirmed only after payment of the reservation fee has been verified with the gateway.
BR-05	A booking may be cancelled without a fee when the cancellation is made at least 30 minutes before its start; a later cancellation incurs the cancellation fee in the tariff.
BR-06	A booking not checked in within the grace period (default 15 minutes) after its start is a No-Show, and the no-show fee applies.
BR-07	The tariff in force when a booking is confirmed applies to that booking, even if the operator later changes the tariff.
BR-08	Auto-booking shall not select a station that is not reachable with the usable range, or a connector that is not compatible.
BR-09	Operators may act only on their own stations; an Administrator approves operators and stations.
BR-10	A driver with an unpaid balance cannot create new bookings until it is settled.
BR-11	An inactive station, charger or connector shall not appear in new recommendations or search results.
BR-12	Analytics shown to a user shall be limited to data that the user's role permits.
BR-13	An idle fee applies when a vehicle stays connected more than 10 minutes after charging has ended.

7. Other Requirements

7.1 Data Retention and Privacy

- Raw location coordinates shall be retained for no longer than 24 hours, consistent with REQ-F1-19.
- Invoices and payment records shall be retained for the period required by applicable financial regulations, with personal data minimised.
- A driver shall be able to request deletion of their account; on deletion, personally identifying data shall be removed while anonymised aggregate analytics may be retained.
- Personal data shall be collected and used only for the purposes described in this SRS, in line with the Digital Personal Data Protection Act, 2023.

7.2 Internationalization

- Version 1.0 supports the English language only.
- Amounts shall be displayed in Indian Rupees (INR), distances in kilometres and energy in kWh.
- Dates and times shall be displayed in the time zone of the device, with Indian Standard Time as the system time zone for slots.

7.3 Legal and Licensing

- Use of the Maps, payment and notification services shall comply with their terms of service and quotas, including required attribution on maps.
- The system shall comply with the payment gateway's requirements so that card data is never handled by the system.
- Prices shown to drivers shall include applicable taxes or shall show them as a separate line before booking.

7.4 Deployment and Installation

- The backend, database and cache shall be containerised with Docker so that the environment can be provisioned reproducibly.
- Builds, tests and deployments shall run through the Jenkins continuous integration and delivery pipeline, which shall also produce the installable app builds.
- Static analysis via SonarQube shall be part of the pipeline, and a build that fails the agreed quality gate shall not be promoted.
- The test environment shall use the charge point simulator and the payment gateway's test mode.

7.5 Future Extensibility

- The rule-based scoring in auto-booking shall be replaceable by a trained model (for example for demand prediction) without redesigning the accounts, discovery or payment subsystems.
- The data model shall permit reading the battery level from the vehicle, dynamic pricing, multi-language support, a web console for operators, and additional user roles.

Appendix A: Glossary

Term	Definition
Administrator	A user role that approves operators and stations, manages users and roles, and sets platform limits.
Audit record	A stored, attributable record of an administrative action such as a role change or tariff change.
Auto-booking	Automatic selection and booking of the best compatible, reachable slot, using a weighted score, without the driver choosing the slot.
Battery level (SoC)	State of charge: the percentage of battery capacity that is currently filled, from 0 to 100.
Booking	A reservation of one slot window on one connector for one driver and vehicle.
Booking reference	A unique code, also shown as a QR code, that identifies a confirmed booking.
Charge point (charger)	A physical unit that delivers electricity to a vehicle. It may have one or more connectors.
Charging mode	The driver's preference that sets the scoring weights: Fastest, Cheapest or Nearest.
Charging session	The period during which a vehicle is actually receiving energy at a connector.
Connector	The plug socket on a charger. Types in version 1.0: CCS2, CHAdeMO, Type 2 and GB/T.
Driver	A user who owns an EV and uses the system to find, book and pay for charging.
Grace period	The time after a booking's start during which the driver may still check in before the booking becomes a No-Show.
Hold	A temporary reservation of a slot, lasting 5 minutes, while the driver completes payment.
Idle fee	A per-minute fee charged when a vehicle stays connected after charging has ended.
OCPP	Open Charge Point Protocol: the standard through which chargers report status and receive commands.
No-Show	A booking where the driver did not check in within the grace period.
RBAC	Role-Based Access Control. Permissions granted according to a user's role.
Reachable station	A station whose road distance is within the usable range of the vehicle.
Rebooking	Automatic replacement of a booking with another slot when the original charger fails or is withdrawn.
Reservation fee	A fee paid when a booking is confirmed and credited against the final bill.
Slot	A 15-minute unit of time on one connector. A booking uses one or more consecutive slots.
Slot dashboard	The screen that shows the live status of each connector and the free and booked slots for a date.
Stale status	A connector status for which no update has arrived within 120 seconds; shown as "Status unknown".
Station	A location with one or more chargers, managed by a Station Operator.
Station Operator	A user role that owns or manages stations and sets their tariffs.
Target level	The battery level the driver wants to reach in a charging session (default 80%).
Tariff	The price rules of a station: price per kWh, time-of-use periods, idle fee, reservation fee, and cancellation and no-show fees.
Time-of-use tariff	A tariff in which the price per kWh depends on the time of day.
Usable range	The distance the vehicle can still travel on the current battery level, after keeping a safety reserve.

Appendix B: Field Layouts and Calculation Rules

Battery levels are percentages from 0 to 100. Connector types: {CCS2, CHAdeMO, Type 2, GB/T}. Current types: {AC, DC}. Connector and booking statuses are given below.

B.1 Calculation Rules

Quantity	Rule
Usable range (km)	$\max(0, (\text{SoC_now} - \text{reserve}) / 100 \times \text{battery_kWh}) / (\text{consumption_kWh_per_100km} / 100)$, where reserve = 10 percentage points
Station reachable	road distance to the station $\leq$ usable range
Energy required (kWh)	$(\text{SoC_target} - \text{SoC_now}) / 100 \times \text{battery_kWh} / 0.90$, where 0.90 is the assumed charging efficiency
Effective power (kW)	$\min(\text{connector power, vehicle maximum rate for the connector's current type})$. For DC charging, power for the part above 80% is taken as 50% of this value
Charging time (min)	energy required / effective power $\times 60$ (with the DC reduction applied)
Booked slot length (min)	charging time $\times 1.10$, rounded up to the next multiple of 15; minimum 15, maximum 240
Final bill	energy delivered $\times$ price per kWh + idle fee + taxes – reservation fee paid
Auto-booking score	$w_{\text{time}} \times \text{time_score} + w_{\text{cost}} \times \text{cost_score} + w_{\text{distance}} \times \text{distance_score} + w_{\text{reliability}} \times \text{reliability_score}$, each score normalised to 0–1 across the candidates (higher is better)

Default auto-booking weights:

Charging mode	w_time	w_cost	w_distance	w_reliability
Fastest	0.50	0.10	0.25	0.15
Cheapest	0.15	0.50	0.20	0.15
Nearest	0.15	0.10	0.60	0.15

B.2 Configurable Defaults

Parameter	Default
Slot length unit	15 minutes
Booking horizon	7 days; minimum lead time 5 minutes
Slot hold duration	5 minutes
Free cancellation	at least 30 minutes before start
No-show grace period	15 minutes
Idle fee grace period	10 minutes
Stale status limit	120 seconds
Low-battery threshold	20%
Default target level	80%
Search radius	10 km (1 to 50 km)
Reservation fee	INR 50 per booking

B.3 Vehicle and User Fields

Field	Length / range	Data Type	Description	Mandatory
user_id	10	Numeric	Unique identifier of the user account	Y
name	60	String	User name	Y
email	120	String	Email, used for login	Y
phone	15	String	Mobile number	Y
role	15	Enum	Driver, Station Operator, Administrator	Y
password_hash	255	String	Securely hashed password (never plaintext)	Y
vehicle_id	10	Numeric	Unique identifier of the vehicle	Y
make, model	40 each	String	Vehicle make and model	Y
battery_kwh	5 – 250	Decimal	Battery capacity in kWh	Y
connector_type	10	Enum	CCS2, CHAdeMO, Type 2, GB/T	Y
max_ac_kw	1 – 50	Decimal	Maximum AC charging rate in kW	Y
max_dc_kw	10 – 400	Decimal	Maximum DC charging rate in kW (0 if none)	N
consumption_kwh_100km	5 – 50	Decimal	Energy consumption in kWh per 100 km	Y
registration_no	15	String	Vehicle registration number	N

B.4 Station, Charger and Slot Fields

Field	Length / range	Data Type	Description	Mandatory
station_id	10	Numeric	Unique identifier of the station	Y
name	100	String	Station name	Y
address	200	String	Street address	Y
lat, lon	–90..90, –180..180	Decimal	Station coordinates	Y
amenities	list	String list	For example restroom, cafe, parking	N
opening_hours	list	Time ranges	Daily opening hours (may be 24 hours)	Y
approval_status	10	Enum	pending, approved, rejected, inactive	Y
charger_id	10	Numeric	Unique identifier of the charger	Y
power_kw	1 – 400	Decimal	Rated power of the charger in kW	Y
current_type	2	Enum	AC or DC	Y
connector_id	10	Numeric	Unique identifier of the connector	Y
connector status	10	Enum	Available, Occupied, Reserved, Faulted, Offline	Y
slot_id	10	Numeric	Unique identifier of the slot	Y
start_time, end_time	—	Timestamp	Slot window in UTC, 15-minute aligned	Y
slot state	10	Enum	Free, Held, Booked, Blocked	Y

B.5 Booking, Session, Payment and Tariff Fields

Field	Length / range	Data Type	Description	Mandatory
booking_id	10	Numeric	Unique identifier of the booking	Y
start_soc, target_soc	0 – 100	Numeric	Battery level at booking and the target level	Y
est_energy_kwh	—	Decimal	Estimated energy required	Y
est_duration_min	15 – 240	Numeric	Booked slot length in minutes	Y
booking_type	10	Enum	manual or auto	Y
booking status	12	Enum	Held, Confirmed, Active, Completed, Cancelled, No-Show, Expired, Reassigned	Y
session_id	10	Numeric	Unique identifier of the charging session	Y
energy_kwh	—	Decimal	Energy delivered in the session	N
payment_id	10	Numeric	Unique identifier of the payment	Y
amount	—	Integer	Amount in paise	Y
payment type	15	Enum	reservation, balance, refund, fee	Y
payment status	10	Enum	success, failure, pending, refunded	Y
gateway_ref	64	String	Reference issued by the payment gateway	N
price_per_kwh	—	Integer	Price per kWh in paise	Y
idle_fee_per_min	—	Integer	Idle fee per minute in paise	N
peak_start, peak_end	—	Time	Time-of-use period of a higher price	N

Appendix C: Requirement Traceability Matrix

Architecture Reference abbreviations: HLD = High Level Architecture Document (to be produced). Design Reference abbreviations: SDD = Software Design Document (to be produced). Code File Reference values are indicative placeholders to be updated when module names are finalised. Test Case IDs follow the convention TC-Fn-nn (unit/integration) and STC-Fn-nn (system test).

Feature 1 — User Accounts, Vehicle Profile and Access Control

Sl. No	Requirement ID	Brief Description	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
1	REQ-F1-01	Register account	HLD \$Auth	SDD \$Auth	auth/views.py	TC-F1-01	STC-F1-01
2	REQ-F1-02	Password policy	HLD \$Auth	SDD \$Auth	auth/validators.py	TC-F1-02	STC-F1-02
3	REQ-F1-03	Verify contact details	HLD \$Auth	SDD \$Auth	auth/verification.py	TC-F1-03	STC-F1-03
4	REQ-F1-04	Login and tokens	HLD \$Auth	SDD \$Auth	auth/views.py	TC-F1-04	STC-F1-04
5	REQ-F1-05	Login lockout	HLD \$Auth	SDD \$Auth	auth/lockout.py	TC-F1-05	STC-F1-05
6	REQ-F1-06	Password reset	HLD \$Auth	SDD \$Auth	auth/reset.py	TC-F1-06	STC-F1-06
7	REQ-F1-07	Logout	HLD \$Auth	SDD \$Auth	auth/views.py	TC-F1-07	STC-F1-07
8	REQ-F1-08	Reject duplicate account	HLD \$Auth	SDD \$Auth	auth/validators.py	TC-F1-08	STC-F1-08
9	REQ-F1-09	Generic login error	HLD \$Auth	SDD \$Auth	auth/views.py	TC-F1-09	STC-F1-09
10	REQ-F1-10	Assign one role	HLD \$Auth	SDD \$Auth	auth/models.py	TC-F1-10	STC-F1-10
11	REQ-F1-11	Server-side RBAC	HLD \$Auth	SDD \$Auth	auth/permissions.py	TC-F1-11	STC-F1-11
12	REQ-F1-12	Operator data scope	HLD \$Auth	SDD \$Auth	auth/permissions.py	TC-F1-12	STC-F1-12
13	REQ-F1-13	Admin user management	HLD \$Auth	SDD \$Auth	admin/users.py	TC-F1-13	STC-F1-13
14	REQ-F1-14	Audit user actions	HLD \$Auth	SDD \$Auth	audit/models.py	TC-F1-14	STC-F1-14
15	REQ-F1-15	Operator onboarding	HLD \$Auth	SDD \$Auth	operators/views.py	TC-F1-15	STC-F1-15
16	REQ-F1-16	Edit profile and preferences	HLD \$Profile	SDD \$Profile	profile/views.py	TC-F1-16	STC-F1-16
17	REQ-F1-17	Profile validation	HLD \$Profile	SDD \$Profile	profile/validators.py	TC-F1-17	STC-F1-17
18	REQ-F1-18	Manual location fallback	HLD \$Profile	SDD \$Profile	profile/location.py	TC-F1-18	STC-F1-18
19	REQ-F1-19	Location retention limit	HLD \$Profile	SDD \$Profile	tasks/cleanup.py	TC-F1-19	STC-F1-19
20	REQ-F1-20	Account deletion	HLD \$Profile	SDD \$Profile	profile/deletion.py	TC-F1-20	STC-F1-20
21	REQ-F1-21	Vehicle CRUD	HLD \$Vehicles	SDD \$Vehicles	vehicles/views.py	TC-F1-21	STC-F1-21
22	REQ-F1-22	Vehicle limit and default	HLD \$Vehicles	SDD \$Vehicles	vehicles/models.py	TC-F1-22	STC-F1-22
23	REQ-F1-23	Preset EV models	HLD \$Vehicles	SDD \$Vehicles	vehicles/presets.py	TC-F1-23	STC-F1-23
24	REQ-F1-24	Vehicle value validation	HLD \$Vehicles	SDD \$Vehicles	vehicles/validators.py	TC-F1-24	STC-F1-24

Sl. No	Requirement ID	Brief Description	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
25	REQ-F1-25	Protect booked vehicle	HLD §Vehicles	SDD §Vehicles	vehicles/validators.py	TC-F1-25	STC-F1-25
26	REQ-F1-26	Driver sees own data only	HLD §Vehicles	SDD §Vehicles	auth/permissions.py	TC-F1-26	STC-F1-26

Feature 2 — Station Discovery, Navigation and Real-Time Slot Dashboard

Sl. No	Requirement ID	Brief Description	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
27	REQ-F2-01	Station CRUD	HLD §Stations	SDD §Stations	stations/views.py	TC-F2-01	STC-F2-01
28	REQ-F2-02	Charger and connector CRUD	HLD §Stations	SDD §Stations	stations/chargers.py	TC-F2-02	STC-F2-02
29	REQ-F2-03	Station approval gate	HLD §Stations	SDD §Stations	stations/moderation.py	TC-F2-03	STC-F2-03
30	REQ-F2-04	Station validation	HLD §Stations	SDD §Stations	stations/validators.py	TC-F2-04	STC-F2-04
31	REQ-F2-05	Out-of-service marking	HLD §Stations	SDD §Stations	stations/chargers.py	TC-F2-05	STC-F2-05
32	REQ-F2-06	Maintenance windows	HLD §Stations	SDD §Stations	stations/slots.py	TC-F2-06	STC-F2-06
33	REQ-F2-07	Nearby station search	HLD §Discovery	SDD §Discovery	discovery/search.py	TC-F2-07	STC-F2-07
34	REQ-F2-08	Availability markers	HLD §Discovery	SDD §Discovery	discovery/views.py	TC-F2-08	STC-F2-08
35	REQ-F2-09	Filter stations	HLD §Discovery	SDD §Discovery	discovery/filters.py	TC-F2-09	STC-F2-09
36	REQ-F2-10	Sort stations	HLD §Discovery	SDD §Discovery	discovery/search.py	TC-F2-10	STC-F2-10
37	REQ-F2-11	Text search	HLD §Discovery	SDD §Discovery	discovery/search.py	TC-F2-11	STC-F2-11
38	REQ-F2-12	Pagination and clustering	HLD §Discovery	SDD §Discovery	discovery/views.py	TC-F2-12	STC-F2-12
39	REQ-F2-13	Station detail	HLD §Discovery	SDD §Discovery	discovery/views.py	TC-F2-13	STC-F2-13
40	REQ-F2-14	Favourite stations	HLD §Discovery	SDD §Discovery	discovery/favourites.py	TC-F2-14	STC-F2-14
41	REQ-F2-15	Rate a station	HLD §Discovery	SDD §Discovery	discovery/reviews.py	TC-F2-15	STC-F2-15
42	REQ-F2-16	Report faulty charger	HLD §Discovery	SDD §Discovery	discovery/reports.py	TC-F2-16	STC-F2-16
43	REQ-F2-17	Ingest connector status	HLD §Dashboard	SDD §Dashboard	gateway/ingest.py	TC-F2-17	STC-F2-17
44	REQ-F2-18	Live status display	HLD §Dashboard	SDD §Dashboard	dashboard/views.py	TC-F2-18	STC-F2-18
45	REQ-F2-19	Live status push	HLD §Dashboard	SDD §Dashboard	dashboard/realtime.py	TC-F2-19	STC-F2-19
46	REQ-F2-20	Stale status handling	HLD §Dashboard	SDD §Dashboard	dashboard/staleness.py	TC-F2-20	STC-F2-20
47	REQ-F2-21	Reserved status	HLD §Dashboard	SDD §Dashboard	dashboard/state.py	TC-F2-21	STC-F2-21
48	REQ-F2-22	Slot generation	HLD §Dashboard	SDD §Dashboard	stations/slots.py	TC-F2-22	STC-F2-22
49	REQ-F2-23	Slot timeline	HLD §Dashboard	SDD §Dashboard	dashboard/timeline.py	TC-F2-23	STC-F2-23
50	REQ-F2-24	Estimated wait time	HLD §Dashboard	SDD §Dashboard	dashboard/wait.py	TC-F2-24	STC-F2-24

Sl. No	Requirement ID	Brief Description	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
51	REQ-F2-25	Route preview	HLD §Navigation	SDD §Navigation	navigation/routes.py	TC-F2-25	STC-F2-25
52	REQ-F2-26	Navigation hand-off	HLD §Navigation	SDD §Navigation	navigation/handoff.py	TC-F2-26	STC-F2-26
53	REQ-F2-27	ETA versus slot check	HLD §Navigation	SDD §Navigation	navigation/eta.py	TC-F2-27	STC-F2-27
54	REQ-F2-28	Route cache and fallback	HLD §Navigation	SDD §Navigation	navigation/cache.py	TC-F2-28	STC-F2-28

Feature 3 — Battery-Aware Booking and Auto-Booking

Sl. No	Requirement ID	Brief Description	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
55	REQ-F3-01	Battery level input	HLD §Booking	SDD §Estimator	booking/battery.py	TC-F3-01	STC-F3-01
56	REQ-F3-02	Battery input validation	HLD §Booking	SDD §Estimator	booking/validators.py	TC-F3-02	STC-F3-02
57	REQ-F3-03	Usable range calculation	HLD §Booking	SDD §Estimator	booking/range.py	TC-F3-03	STC-F3-03
58	REQ-F3-04	Reachability filter	HLD §Booking	SDD §Estimator	booking/range.py	TC-F3-04	STC-F3-04
59	REQ-F3-05	Connector compatibility	HLD §Booking	SDD §Estimator	booking/compat.py	TC-F3-05	STC-F3-05
60	REQ-F3-06	Energy estimate	HLD §Booking	SDD §Estimator	booking/estimator.py	TC-F3-06	STC-F3-06
61	REQ-F3-07	Charging time estimate	HLD §Booking	SDD §Estimator	booking/estimator.py	TC-F3-07	STC-F3-07
62	REQ-F3-08	Slot length rule	HLD §Booking	SDD §Estimator	booking/estimator.py	TC-F3-08	STC-F3-08
63	REQ-F3-09	Cost estimate	HLD §Booking	SDD §Estimator	booking/pricing.py	TC-F3-09	STC-F3-09
64	REQ-F3-10	Recommended slots	HLD §Booking	SDD §Booking	booking/recommender.py	TC-F3-10	STC-F3-10
65	REQ-F3-11	Manual slot choice	HLD §Booking	SDD §Booking	booking/views.py	TC-F3-11	STC-F3-11
66	REQ-F3-12	Slot hold	HLD §Booking	SDD §Booking	booking/hold.py	TC-F3-12	STC-F3-12
67	REQ-F3-13	No double booking	HLD §Booking	SDD §Booking	booking/concurrency.py	TC-F3-13	STC-F3-13
68	REQ-F3-14	Driver overlap rule	HLD §Booking	SDD §Booking	booking/validators.py	TC-F3-14	STC-F3-14
69	REQ-F3-15	Confirm booking	HLD §Booking	SDD §Booking	booking/views.py	TC-F3-15	STC-F3-15
70	REQ-F3-16	Auto-book action	HLD §AutoBook	SDD §AutoBook	autobook/engine.py	TC-F3-16	STC-F3-16
71	REQ-F3-17	Weighted slot scoring	HLD §AutoBook	SDD §AutoBook	autobook/scoring.py	TC-F3-17	STC-F3-17
72	REQ-F3-18	Explain the choice	HLD §AutoBook	SDD §AutoBook	autobook/explain.py	TC-F3-18	STC-F3-18
73	REQ-F3-19	Confirmation modes	HLD §AutoBook	SDD §AutoBook	autobook/modes.py	TC-F3-19	STC-F3-19
74	REQ-F3-20	Low-battery trigger	HLD §AutoBook	SDD §AutoBook	autobook/triggers.py	TC-F3-20	STC-F3-20
75	REQ-F3-21	No-slot handling	HLD §AutoBook	SDD §AutoBook	autobook/fallback.py	TC-F3-21	STC-F3-21
76	REQ-F3-22	Booking state machine	HLD §Booking	SDD §Lifecycle	booking/state.py	TC-F3-22	STC-F3-22

Sl. No	Requirement ID	Brief Description	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
77	REQ-F3-23	Modify booking	HLD §Booking	SDD §Lifecycle	booking/views.py	TC-F3-23	STC-F3-23
78	REQ-F3-24	Cancel booking	HLD §Booking	SDD §Lifecycle	booking/cancel.py	TC-F3-24	STC-F3-24
79	REQ-F3-25	No-show handling	HLD §Booking	SDD §Lifecycle	booking/noshow.py	TC-F3-25	STC-F3-25
80	REQ-F3-26	Auto-rebooking	HLD §Booking	SDD §Lifecycle	booking/rebook.py	TC-F3-26	STC-F3-26
81	REQ-F3-27	Check-in	HLD §Session	SDD §Session	session/checkin.py	TC-F3-27	STC-F3-27
82	REQ-F3-28	Remote start	HLD §Session	SDD §Session	session/control.py	TC-F3-28	STC-F3-28
83	REQ-F3-29	Live session monitor	HLD §Session	SDD §Session	session/monitor.py	TC-F3-29	STC-F3-29
84	REQ-F3-30	Stop session	HLD §Session	SDD §Session	session/control.py	TC-F3-30	STC-F3-30
85	REQ-F3-31	Start failure handling	HLD §Session	SDD §Session	session/control.py	TC-F3-31	STC-F3-31
86	REQ-F3-32	Idle fee	HLD §Session	SDD §Session	session/idle.py	TC-F3-32	STC-F3-32
87	REQ-F3-33	Booking notifications	HLD §Session	SDD §Session	notifications/sender.py	TC-F3-33	STC-F3-33

Feature 4 — Payments, Billing and Usage Analytics

Sl. No	Requirement ID	Brief Description	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
88	REQ-F4-01	Payment methods	HLD §Payments	SDD §Payments	payments/views.py	TC-F4-01	STC-F4-01
89	REQ-F4-02	Tokenised payment methods	HLD §Payments	SDD §Payments	payments/tokens.py	TC-F4-02	STC-F4-02
90	REQ-F4-03	Remove payment method	HLD §Payments	SDD §Payments	payments/tokens.py	TC-F4-03	STC-F4-03
91	REQ-F4-04	Reservation fee	HLD §Billing	SDD §Billing	payments/fees.py	TC-F4-04	STC-F4-04
92	REQ-F4-05	Final bill computation	HLD §Billing	SDD §Billing	billing/calc.py	TC-F4-05	STC-F4-05
93	REQ-F4-06	Tariff snapshot	HLD §Billing	SDD §Billing	billing/tariff.py	TC-F4-06	STC-F4-06
94	REQ-F4-07	Settle balance	HLD §Billing	SDD §Billing	billing/settle.py	TC-F4-07	STC-F4-07
95	REQ-F4-08	Refunds	HLD §Billing	SDD §Billing	payments/refunds.py	TC-F4-08	STC-F4-08
96	REQ-F4-09	Payment records	HLD §Billing	SDD §Billing	payments/models.py	TC-F4-09	STC-F4-09
97	REQ-F4-10	Idempotent retry	HLD §Billing	SDD §Billing	payments/idempotency.py	TC-F4-10	STC-F4-10
98	REQ-F4-11	Webhook verification	HLD §Billing	SDD §Billing	payments/webhooks.py	TC-F4-11	STC-F4-11
99	REQ-F4-12	Server-side verification	HLD §Billing	SDD §Billing	payments/verify.py	TC-F4-12	STC-F4-12
100	REQ-F4-13	Daily reconciliation	HLD §Billing	SDD §Billing	payments/reconcile.py	TC-F4-13	STC-F4-13
101	REQ-F4-14	Invoice generation	HLD §Billing	SDD §Invoices	billing/invoice.py	TC-F4-14	STC-F4-14
102	REQ-F4-15	Payment history	HLD §Billing	SDD §Invoices	billing/views.py	TC-F4-15	STC-F4-15

Sl. No	Requirement ID	Brief Description	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
103	REQ-F4-16	Tariff management	HLD §Billing	SDD §Invoices	billing/tariffs.py	TC-F4-16	STC-F4-16
104	REQ-F4-17	Operator earnings	HLD §Billing	SDD §Invoices	billing/earnings.py	TC-F4-17	STC-F4-17
105	REQ-F4-18	Driver usage summary	HLD §Analytics	SDD §Analytics	analytics/driver.py	TC-F4-18	STC-F4-18
106	REQ-F4-19	Driver trend charts	HLD §Analytics	SDD §Analytics	analytics/driver.py	TC-F4-19	STC-F4-19
107	REQ-F4-20	Top stations and CO2	HLD §Analytics	SDD §Analytics	analytics/driver.py	TC-F4-20	STC-F4-20
108	REQ-F4-21	Operator analytics	HLD §Analytics	SDD §Analytics	analytics/operator.py	TC-F4-21	STC-F4-21
109	REQ-F4-22	Platform analytics	HLD §Analytics	SDD §Analytics	analytics/admin.py	TC-F4-22	STC-F4-22
110	REQ-F4-23	Date range filter	HLD §Analytics	SDD §Analytics	analytics/filters.py	TC-F4-23	STC-F4-23
111	REQ-F4-24	CSV export	HLD §Analytics	SDD §Analytics	analytics/export.py	TC-F4-24	STC-F4-24
112	REQ-F4-25	Role-scoped analytics	HLD §Analytics	SDD §Analytics	analytics/permissions.py	TC-F4-25	STC-F4-25
113	REQ-F4-26	Analytics refresh	HLD §Analytics	SDD §Analytics	analytics/aggregator.py	TC-F4-26	STC-F4-26